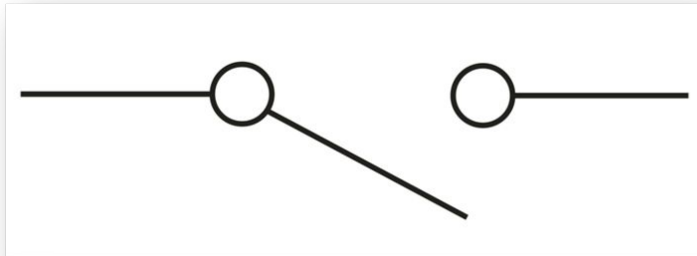
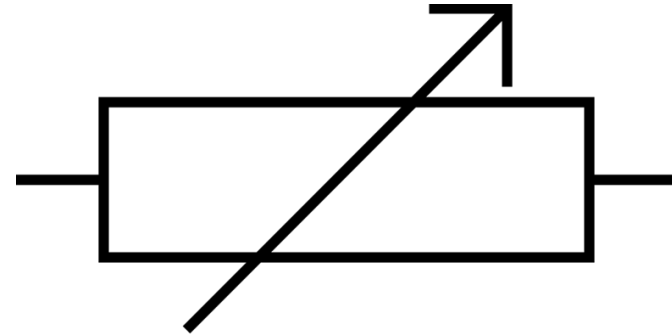
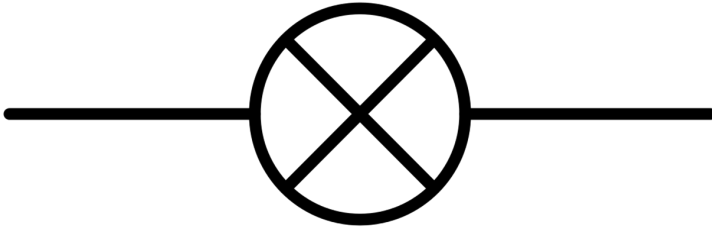
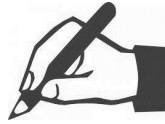


Identify the circuit symbols



DO NOW

Control circuits

Learning Objective:

1. Use a thermistor and Light dependent resistor(LDR)
2. Investigate properties of thermistors, LDRs and Diodes

Keywords

Light Dependent
Resistor

Diode

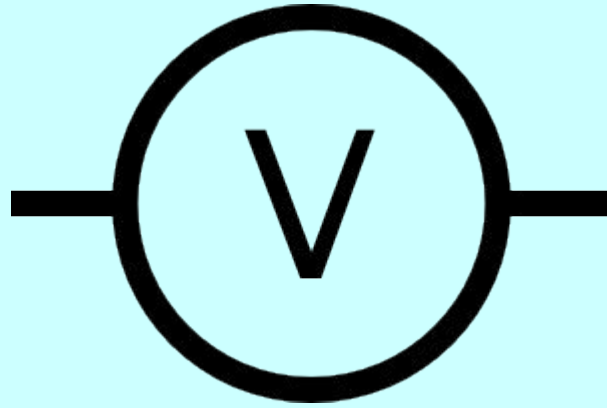
Sensors

Thermistor



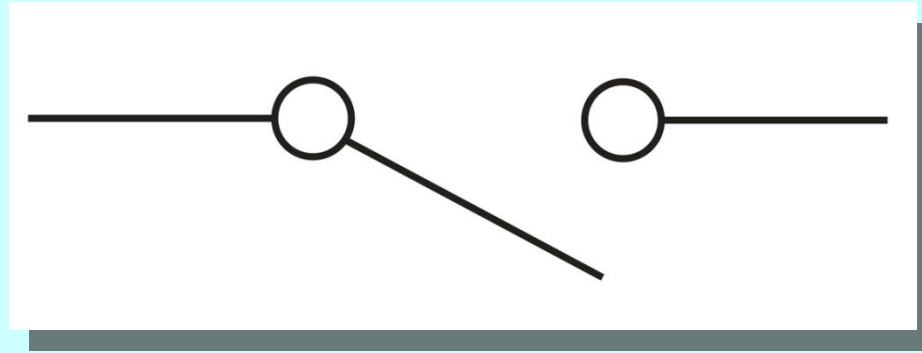
Queen
Elizabeth
School

Which Symbol?



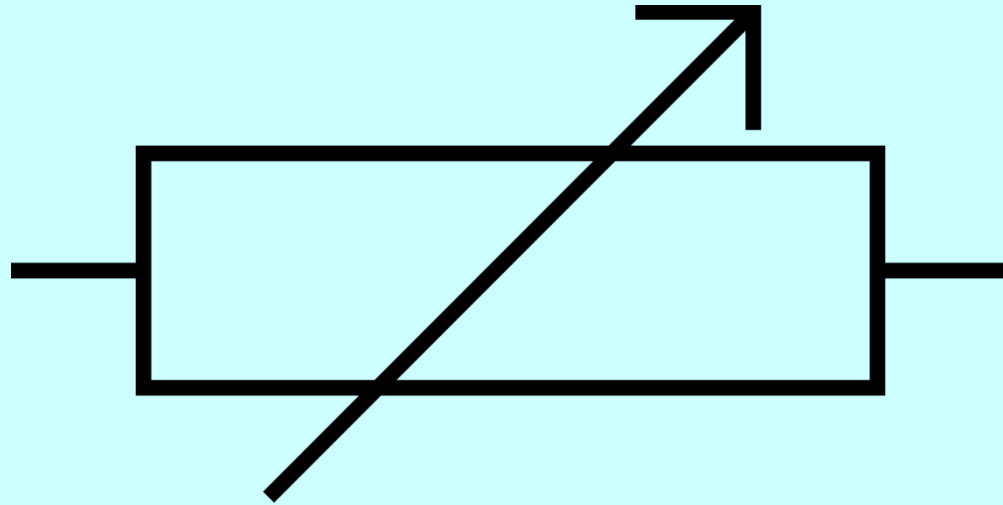
Voltmeter

Which Symbol?



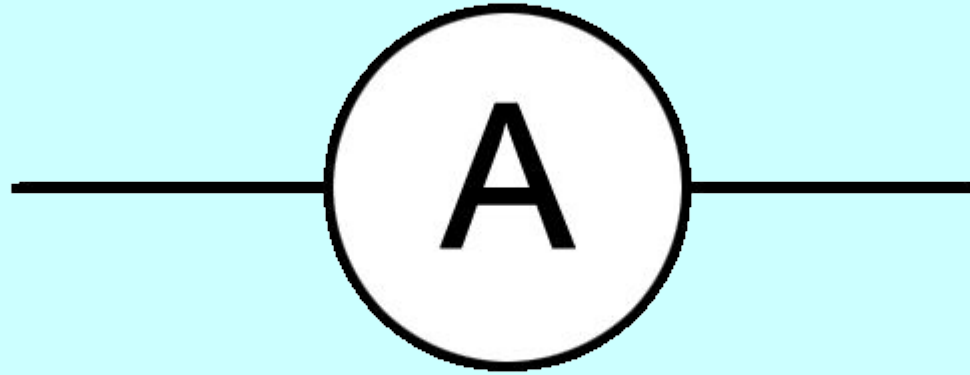
Open switch

Which Symbol?



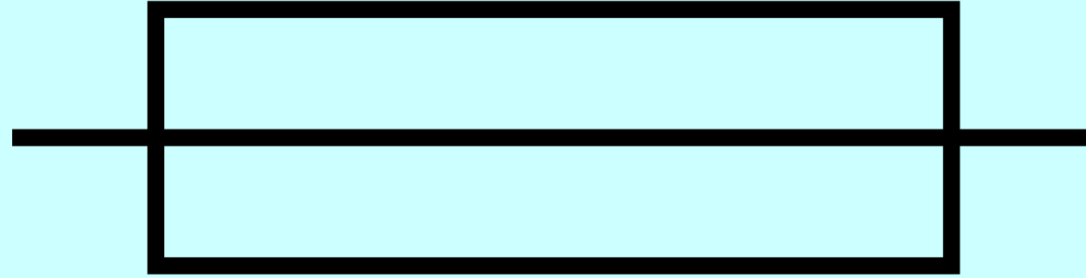
Variable resistor

Which Symbol?



Ammeter

Which Symbol?



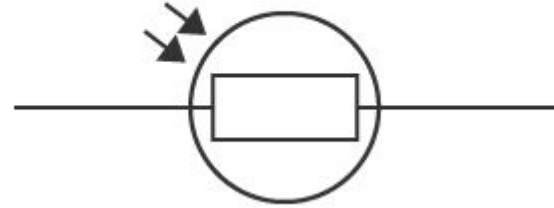
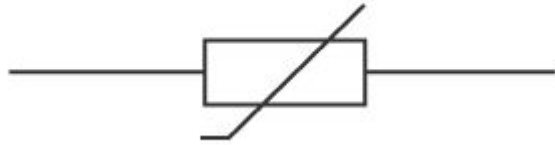
Fuse

Which Symbol?



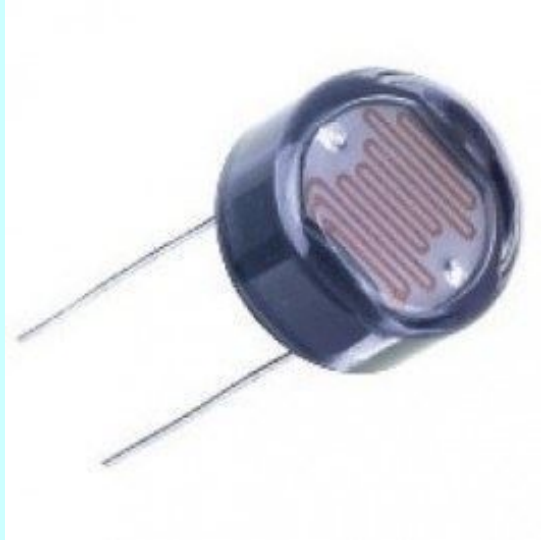
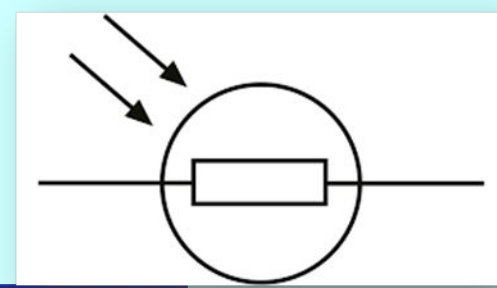
Battery of 2 cells

In your book draw the symbol for a thermistor and an LDR.



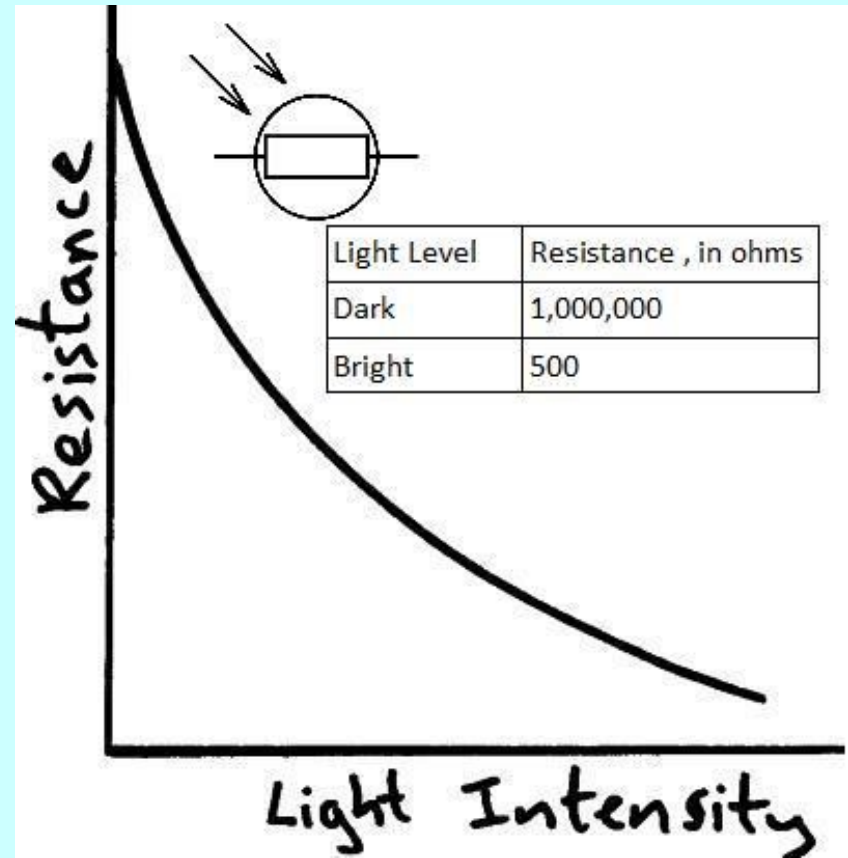
You should be able to know how the resistance of these components can be changed.

What are LDRs?



- LDRs are **Light-dependent resistors** and are often placed on top of street lights to turn them on when it gets **dark**.
-
- Control circuits use components to detect **changes**.
-
- These components are called **sensors**.

Discussion



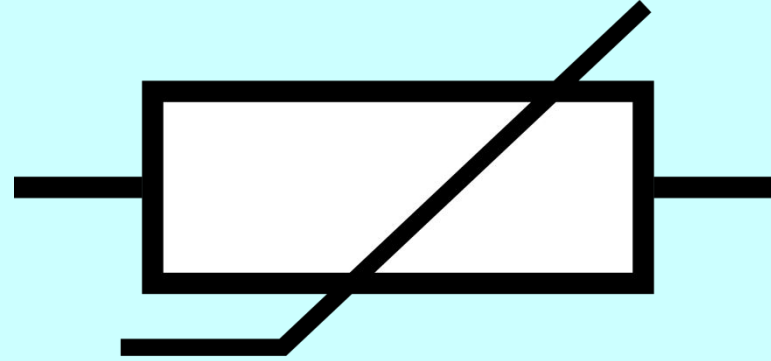
Describe what happens to resistance when light intensity is increased for LDR?

Is a LDR an ohmic conductor?

Explain what happens to current when the light intensity increases?

- In an LDR as light intensity increases what happens to the resistance?
-
- Their resistance decreases as the light intensity increases.
-
- In the dark and at low light levels, the resistance of an LDR is low, and more current can flow through it.
-
- In bright light, the resistance of an LDR is high, and little current can flow through it.

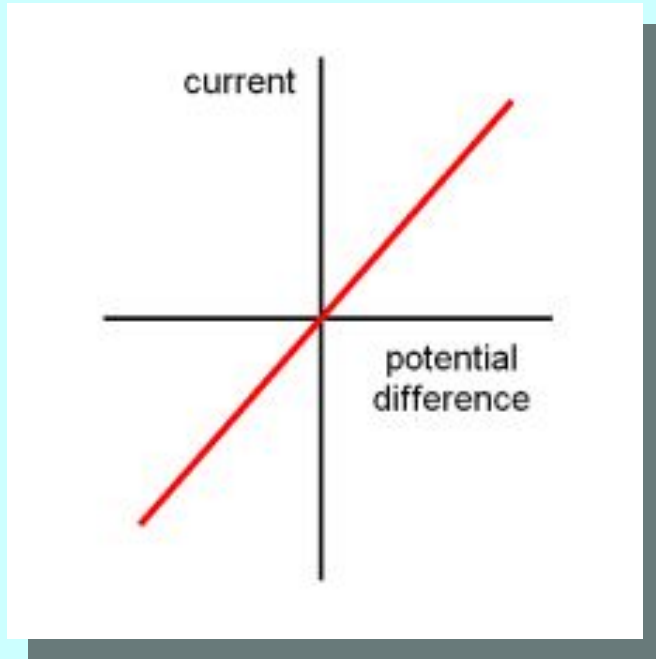
What are thermistors?



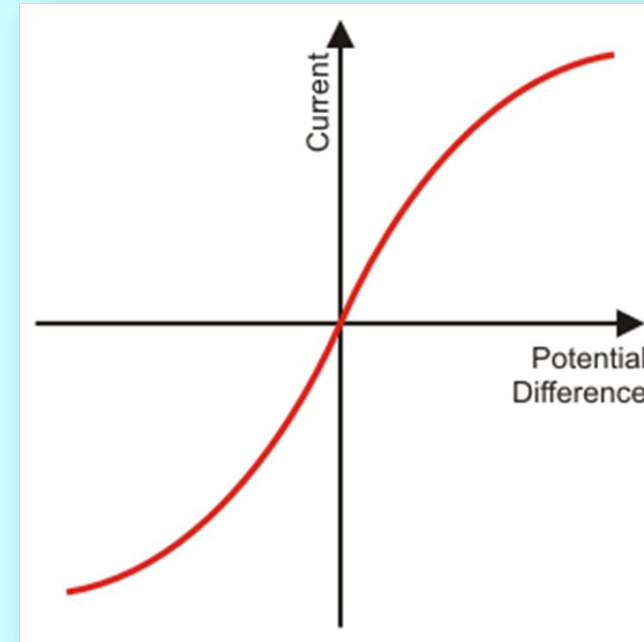
- **Thermistors** is a temperature-dependent resistor.
-
- Its **resistance** changes a lot as **temperature** changes.
-
- At **low** temperatures its resistance is **high** and as the temperature increases its resistance decreases.

- Thermistors are used as temperature sensors, for example, in fire alarms. Their resistance decreases as the temperature increases.
-
- At temperatures, the of a thermistor is, and little current canthrough them.
-
- At temperatures, the resistance of a thermistor is ..., and
-
- At low temperatures, the resistance of a thermistor is high, and little current can flow through them.
-
- At high temperatures, the resistance of a thermistor is low, and more current can flow through them.
-
-

I-V Graphs



Fixed Resistor



Filament bulb

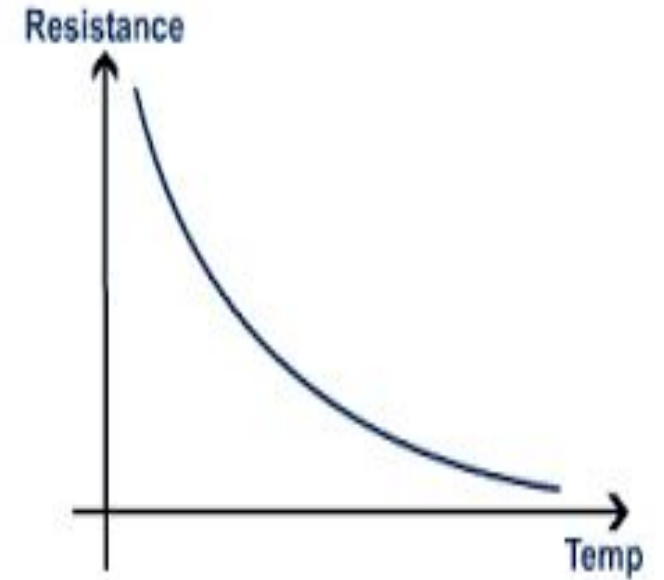
Which one is an ohmic conductor?

Explain why you think it is ohmic conductor?

Explain what happens to resistance when potential difference is increased?

Thermistors continued

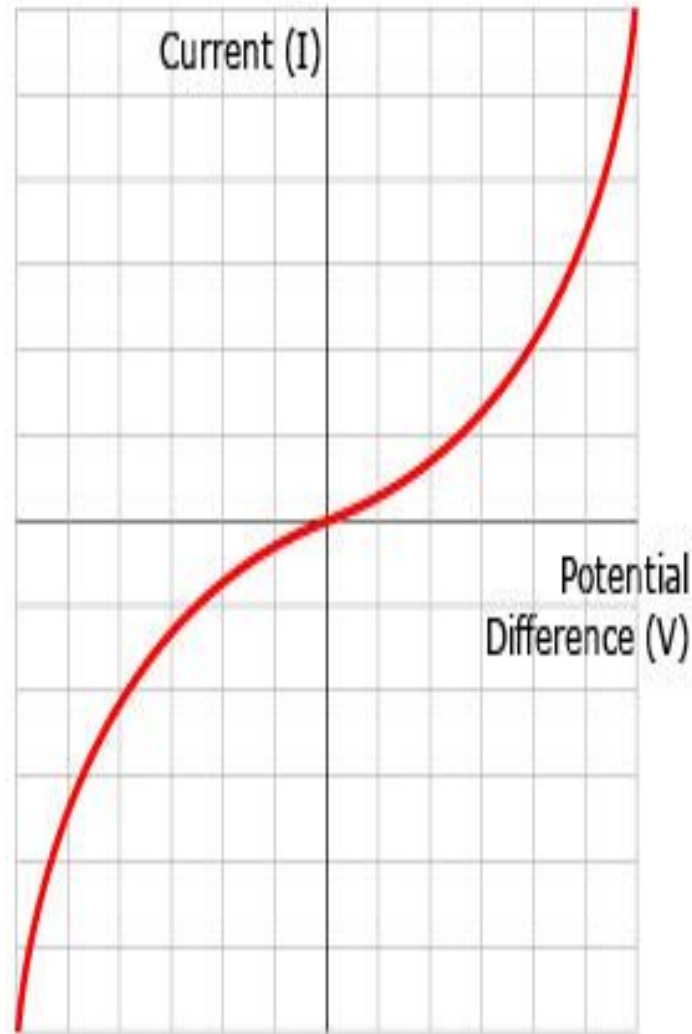
- It is made from a semi-conductor. A semi-conductor is neither a good conductor nor an insulator. When a semi conductor is heated it conducts easily. This is because the rise in temperature releases more free electrons to carry the current. The higher the temperature, the lower the resistance



Thermistor

- The resistance of a thermistor is higher when cold
- **USES:**
- As a fire alarm
- To keep a fish tank from becoming too cold
- To turn on a heater when it gets cold, either in the house or in a green house.

I-V graph of a thermistor



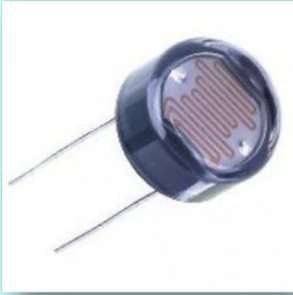
The graph for a thermistor is not a straight line and therefore does not obey Ohm's Law. The plot is actually the reverse of that of the lamp. As more current flows the graph gets steeper. The thermistor is a semiconductor and conducts more electricity when heated. This is because as the temperature increases the thermistor makes available more free electrons to carry the current. Therefore as the current increases the thermistor gets hotter releasing more electrons resulting in a reduction in resistance.

Properties of LDR'S and thermistors

- LDR'S

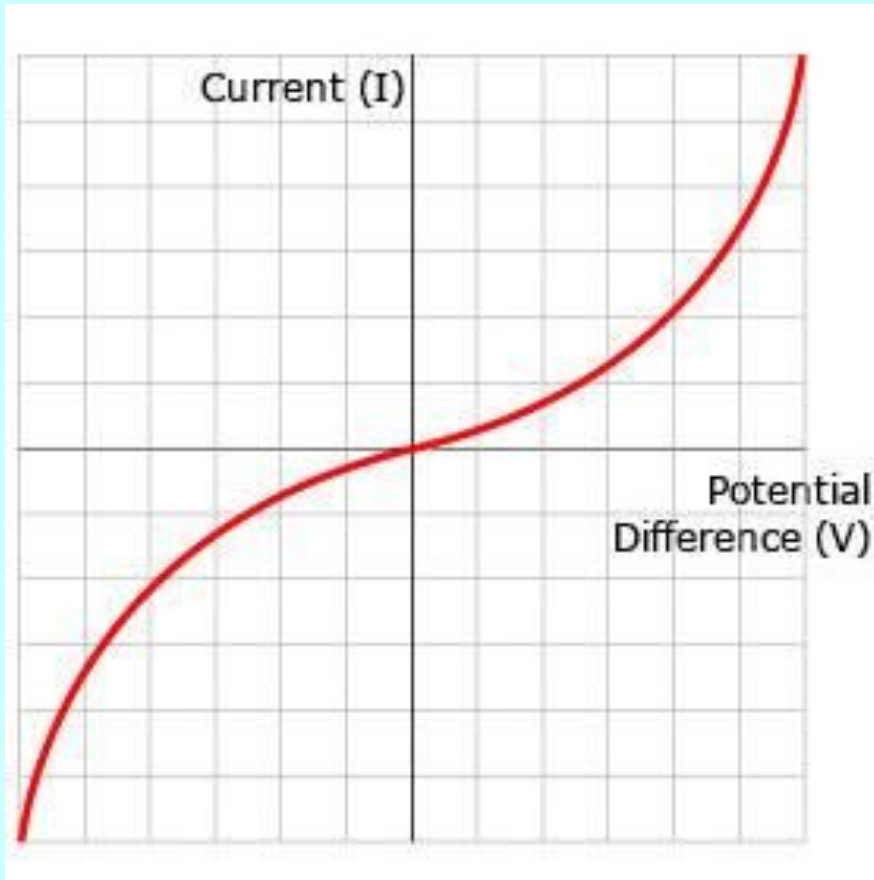
- **Thermistors**

Properties of LDRs



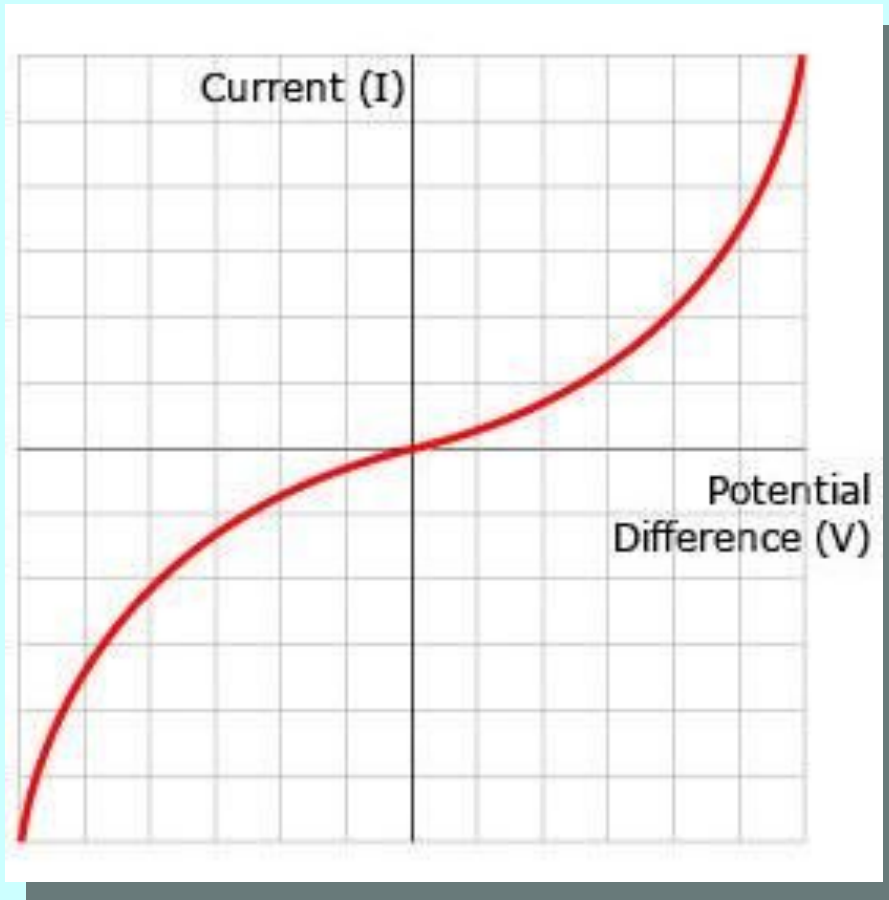
- In bright sunlight an LDR has a resistance of about 100Ω . When it is dark, the resistance of the LDR becomes very large and **over $10,000,000\Omega$** .
-
- An LDR connected in series with a battery and an ammeter can be used to make a **simple light meter**.
-
- When it is bright, the resistance is low (current is high) and when it is dark, the resistance is high (current is low).
-

Properties of Thermistors



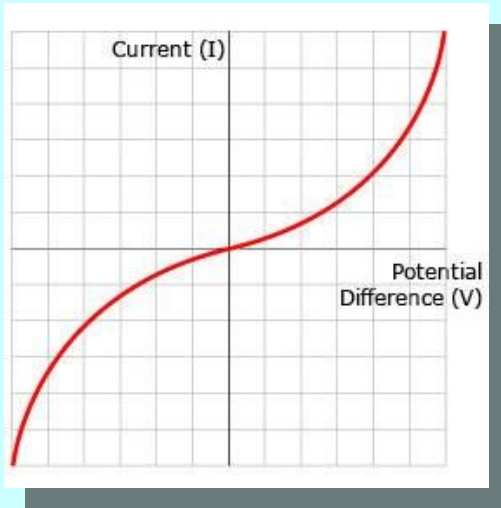
- The resistance of a thermistor decreases as the temperature increases.
- Each thermistor has its own characteristics, but the resistance of a typical thermistor changes from 2000Ω at -20°C to 200Ω at $+20^{\circ}\text{C}$.
-
-

Properties of Thermistors



- A thermistor is made from a **semi-conductor**.
- A semi-conductor is neither a good conductor nor an insulator.
- When a semi-conductor is heated it can **conduct more easily**.
- This is because the rise in temperature releases more free electrons to carry current.
- **The higher the temperature, the lower the resistance.**

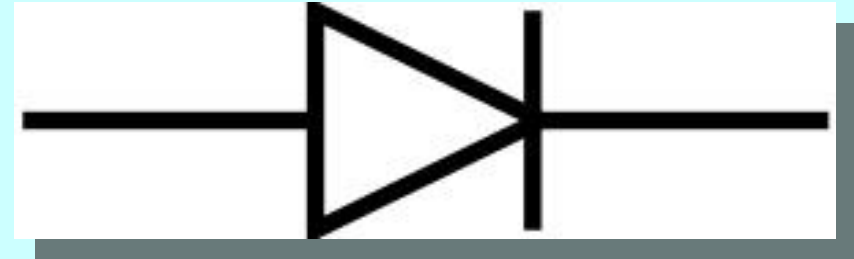
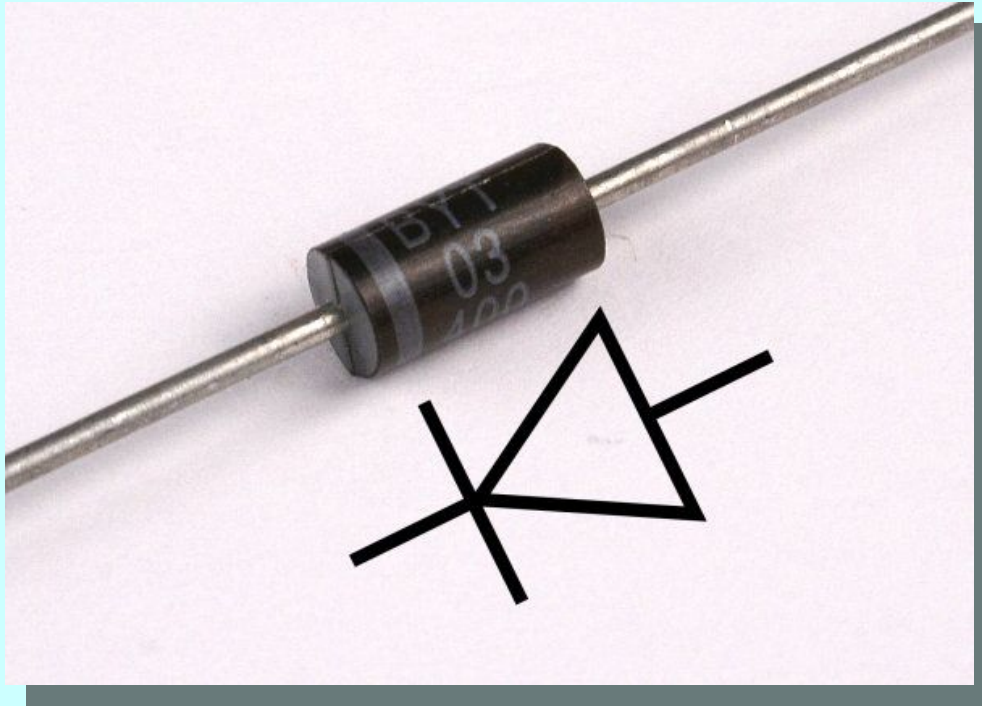
Properties of Thermistors



-
- The higher the temperature, the lower the resistance.
-
- Can be used to:
 1. Turn on a heater when it gets cold;
 2. Act as a fire alarm;
 3. Keep a fish tank from becoming too cold.

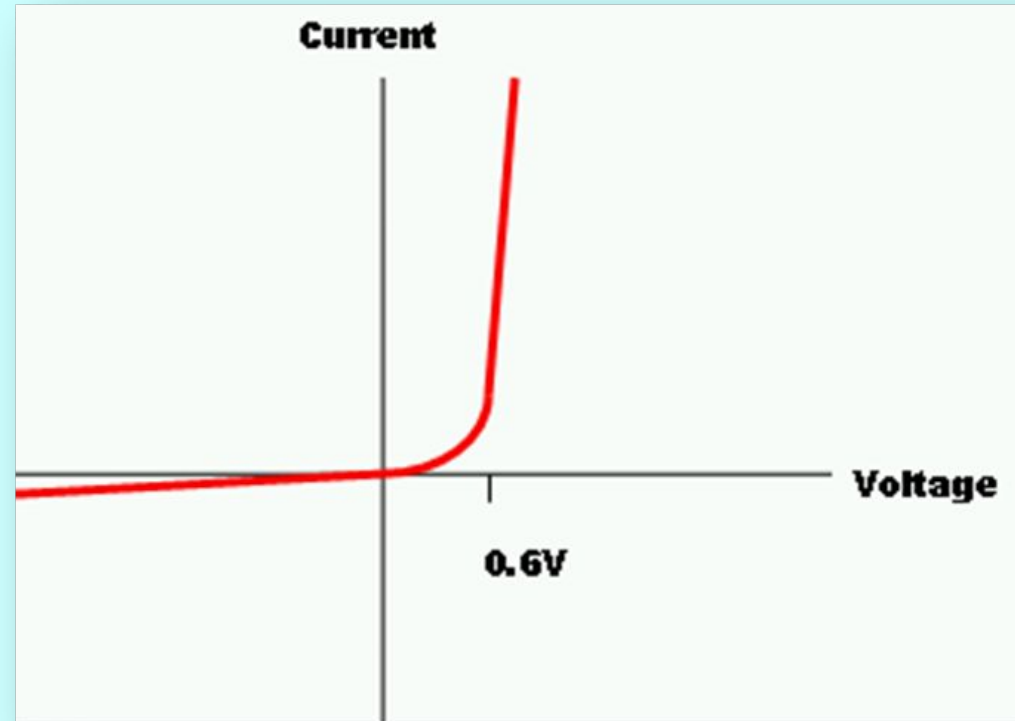


What are diodes?



- A diode is a component that only allows a current to flow in one direction - the direction in which the arrow points in the circuit symbol.
-

Discussion

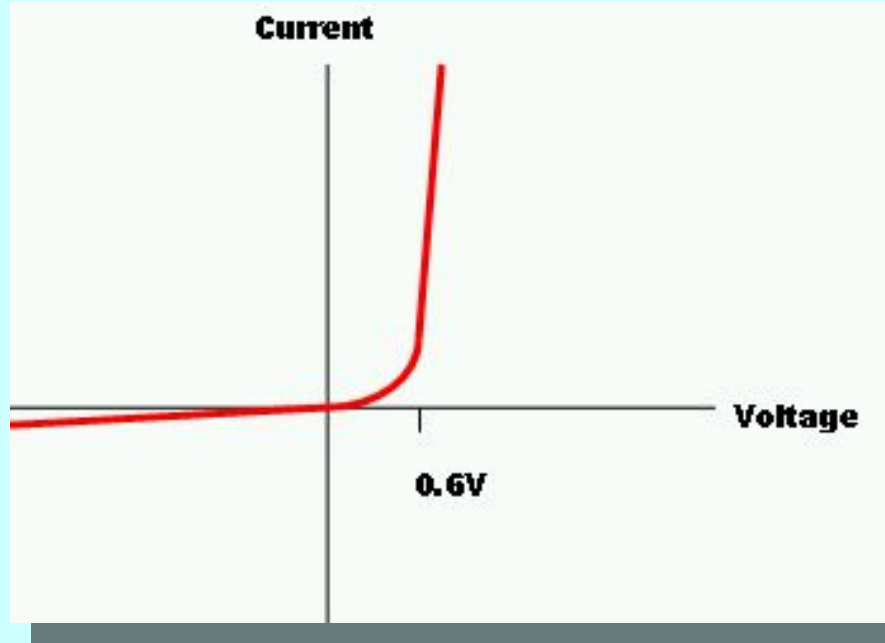


Describe what happens to current when potential difference (voltage) is increased for a diode?

Is a diode an ohmic conductor? Explain?

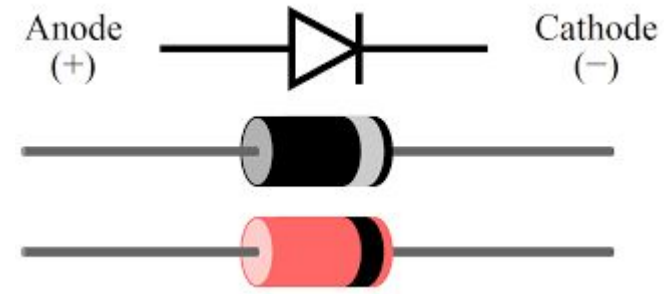
Explain what happens to resistance when p.d. (voltage) increases?

Properties of Diodes

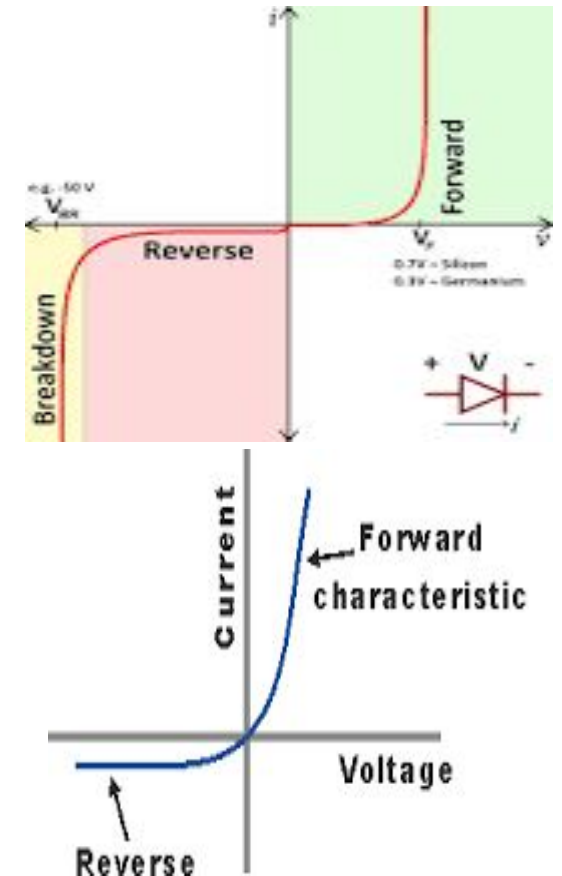


- Most diodes start to conduct when the potential difference across them is about **+0.6V**.
-
- The **steep slope** of the graph shows that the current passes easily through the diode when it is conducting.
-
- With a negative potential difference, very little current flows as resistance is very high.

The properties of a Diode



- A component that only allows a current to flow in one direction (direction in which the arrow shows in the circuit diagram).
- I-V characteristic of a diode can be found using a circuit similar to the one for a filament lamp.
- Most diodes start to conduct when pd across them is about 0.6V
- The steep slope of the graph shows that current passes easily through a diode when it is conducting.
- With a negative pd, very little current flows.



Your turn

- ALL- Answer questions 1-5 p 64-65
- Higher- Answer questions 1-7 from p 64-65